

Subject Description Form

Subject Code	AAE3010
Subject Title	Airline Operations
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite: AAE2004 Introduction to Aviation System and Air Transport Regulation
Objectives	This subject will provide students with 1. The understanding of political, economic, and operational principles and regulations of airline operations; 2. The ability of problem formulation and mathematical modelling for airline operations; and 3. The concept and management style in achieving organisational and operations efficiency in the airline business;
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Formulate mathematical models and deduce their solutions to airline operations related problems; b. Design and validate proper solutions and alternatives in fostering airline business and operations efficiency; c. Design and suggest solutions to ICAO requirements related to cabin safety, cabin crew, pilot flight time limitations, fleet operations and maintenance requirements; and d. Determine airline solutions contributing to the passengers, organisational, societal, economic, and global environment factors.
Subject Synopsis/ Indicative Syllabus	Airline business environment – Policy and regulations in the airline business, market structure and competition; benefits and environmental impacts of air transport; characteristics and risks in the airline industry. Airline financial management – Airline pricing and revenue management; airport slot coordination, policy, and regulation; aircraft leasing and financial operations. Airline operations and management – Airline demand forecasting and analyses, Airline fleet management, airline route planning, Airline competitions and market analyses; regulations in airline operation; Airline ground service management; Human resource management: crew pairing, and rostering management.

Teaching/Learning Methodology	Teaching is conducted through class lectures and projects. The basic knowledge, research methodology and theoretical models will be introduced. The understanding of how to address and formulate problems by using mathematical modelling and operations research is emphasised. Research methodology, case studies, and data analytics skills are taught in class as well as the related real-life scenarios using data to enhance their research abilities.						
	Teaching/Learning Methodology		Intended subject learning outcomes to be covered				
			a	b	c	d	
	1. Lecture		✓	✓	✓	✓	
	2. Project		✓	✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks		% weighting	Intended subject learning outcomes to be assessed			
				a	b	c	d
	1. Assignment		20%	✓	✓	✓	
	2. Group project		30%		✓	✓	✓
	3. Final examination		50%	✓	✓	✓	✓
	Total		100 %				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment The continuous assessment (50%) is aimed at enhancing the students' comprehension and assimilation of various topics of the syllabus via several assignments and group projects. The final examination assessment (50%) will also be considered to assess the students' learning outcomes.						

Student Study Effort Expected	Class contact:	
	▪ Lecture/Project	36 Hrs.
	Other student study effort:	
	▪ Week 7 reading week	3 Hrs.
	▪ Self-study / preparation	39 Hrs.
	▪ Case study, assignment and group project	39 Hrs.
	Total student study effort	117 Hrs.
Reading List and References	1. Abdelghany, A., & Abdelghany, K. (2016). Modeling applications in the airline industry. Routledge. 2. Bazargan, M. (2016). Airline operations and scheduling. Routledge. 3. Clark, P. (2017). Buying the big jets: fleet planning for airlines. Taylor & Francis. 4. Wu, C.-L. (2016). Airline operations and delay management: insights from airline economics, networks and strategic schedule planning: Routledge.	

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